

Research Article

Long Short-Term Memory (LSTM) Based Model for Forecasting of Students' Dropout

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Received: 27/May/2025; Accepted: 28/Jun/2025; Published: 31/Jul/2025

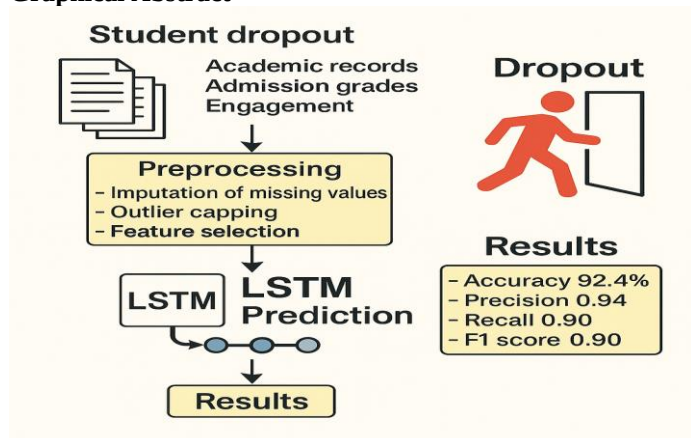


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Abstract— This study addresses the critical issue of student dropout in higher education by proposing a Long Short-Term Memory (LSTM) network to enhance predictive accuracy through sequential data analysis. Traditional models like RandomForest, K-Nearest Neighbors (KNN), and Deep Neural Networks (DNN) often fail to capture temporal dependencies in student records, limiting their effectiveness for early intervention. The proposed LSTM model processes academic records, admission grades, and engagement metrics (e.g., Moodle activity) after preprocessing steps including imputation of missing values, outlier capping via interquartile range (IQR), correlation-based feature selection, and Min-Max normalization. The architecture employs ReLU, sigmoid, and tanh activation functions across three hidden layers, optimized using Bayesian regularization backpropagation. Evaluated against benchmark models, the LSTM achieved 92.4% accuracy, 0.94 precision, 0.90 recall, and 0.90 F1 score, outperforming Random Forest (85.3%), KNN (81.2%), and DNN (88.7%) while reducing training time by 85.6% with a 72-epoch convergence rate. Key predictors identified include admission grades, accumulated credits, and Moodle engagement. Visualizations such as boxplots and correlation matrices highlighted skewed grade distributions and feature interdependencies. The study underscores LSTM's superiority in modeling sequential data for dropout prediction, offering institutions actionable insights to support at-risk students. Future work should address scalability, data privacy, and integration of external factors like teaching quality. This research advocates for AI-driven solutions in educational analytics to improve retention and academic success globally.

Keywords— Long Short-Term Memory (Lstm), Based Model, Forecasting, Students' Dropout

Graphical Abstract-



Purpose- The purpose of a graphical abstract is to visually summarize the key components of a research study in a concise and easily digestible format. Unlike traditional text-based abstracts, a

graphical abstract uses images, icons, and flowcharts to present the study's objective, methodology, major findings, and implications. This visual representation enhances the accessibility of the research, allowing readers to quickly grasp the core message without needing to read the full paper. It is especially useful for busy professionals, researchers, and policymakers who seek to identify relevant studies at a glance. Additionally, graphical abstracts improve the visibility and shareability of research, particularly on digital platforms and social media, where visual content tends to attract more attention. By supporting quicker understanding and better memory retention, graphical abstracts serve as effective tools for scientific communication and dissemination, complementing the written abstract and reinforcing the impact of the study.

1. Introduction

Artificial Intelligence (AI) has significantly transformed various sectors, including education. It plays a critical role in providing insights into students' learning behaviors and enhancing academic outcomes. These innovations offer

viable solutions to some of the key challenges confronting the education sector, especially the persistent problem of underperformance in higher institutions. Tertiary education is grappling with an ongoing challenge of high dropout rates, despite increased investments and supportive initiatives. This situation not only diminishes individual prospects, limiting their future earnings and career paths, but also poses broader societal setbacks by curbing potential contributions to economic development and innovation. AI stands out as a promising remedy to this issue. By employing tools such as Deep Learning, Automation, and Natural Language Processing (NLP), AI facilitates a deeper understanding of student learning processes and supports strategies for academic improvement.

Deep Learning Algorithms: These models analyze extensive educational datasets including academic results, demographic information, and online engagement logs to identify patterns and predict student performance accurately. For instance, such systems can flag students at risk of failure early enough to implement timely interventions [6]. **Automation:** Automated platforms customize learning experiences by adapting content to suit individual learner profiles and providing immediate feedback. This not only streamlines grading processes but also enables instructors to dedicate more time to personalized mentorship and guidance [1]. **Natural Language Processing (NLP):** AI-powered tutoring systems that utilize NLP are capable of engaging students in human-like dialogue, answering questions, giving academic direction, and even offering emotional support. These features contribute to a more personalized and interactive learning environment [2].

Academic underperformance remains a global concern in universities, often reflected in high failure rates and alarming dropout statistics. This problem arises from a variety of interconnected factors, including family background, psychological attributes, teaching techniques, and cognitive abilities [3]. Poor academic performance not only obstructs student progress but can also damage a university's reputation, since academic metrics are commonly used to evaluate institutional effectiveness. Students who consistently underperform are more likely to fail repeatedly and eventually withdraw from their programs [4]. The situation was further exacerbated during the COVID-19 pandemic, which led to the rapid adoption of hybrid learning formats. While these formats offer flexibility and opportunities for skill enhancement, they also introduce new obstacles, such as increased distractions and decreased student engagement [5]. In response to these challenges, AI-based predictive models have emerged as effective tools. Studies by [6] and [7] have demonstrated the use of techniques such as K-Nearest Neighbors (KNN), Random Forest, and Deep Neural Networks (DNN) to forecast student outcomes, achieving prediction accuracies between 78% and 80%. However, many of these studies fall short in demonstrating how their findings can be practically applied to benefit both students and academic institutions. The Long Short-Term Memory (LSTM) network is a specialized form of deep neural network designed to handle sequential data and identify long- and

short-term patterns over time. This makes it particularly suitable for tasks involving the prediction of complex and nonlinear time-series trends [8]. LSTM networks differ from standard Recurrent Neural Networks (RNNs) through their unique gating mechanisms, comprising: **Forget Gate:** Determines which previous information should be discarded, allowing the model to ignore irrelevant or outdated data.

Input Gate: Regulates the inclusion of new information into the memory state, maintaining a balance between fresh input and historical context. **Output Gate:** Controls which parts of the memory should be output, dynamically adjusting focus based on the most relevant features of the data sequence. These mechanisms enable LSTMs to avoid common issues such as vanishing or exploding gradients, which typically hinder the learning of long-term dependencies. As a result, LSTMs are highly effective for applications like predicting student dropout trends, where understanding patterns over time is essential. Advanced LSTM-based models, such as the Attention-aware Stacked Bidirectional LSTM (ASIST), have made notable strides in educational predictive modeling. Utilizing the temporal processing strengths of LSTMs, ASIST analyzes student behavior patterns such as weekly and daily activity in Virtual Learning Environments (VLEs) to classify learners into high, moderate, or at-risk performance groups. It has achieved commendable Area Under the Curve (AUC) scores ranging from 0.86 to 0.90 [2].

The implementation of ASIST highlights the effectiveness of LSTM in capturing complex behavioral dynamics, yielding insights that support proactive intervention strategies. These insights enable educators to customize support for individual students, improving retention rates and overall academic performance. Consequently, LSTM-driven systems like ASIST underscore the transformative role of AI technologies in academic enhancement. This study seeks to address the identified gaps by developing an LSTM-based predictive model for forecasting academic performance and identifying students at risk of dropping out. In contrast to traditional models, the proposed approach leverages the sequential nature of educational data and excels in time-series prediction. Through extensive educational datasets, the model aims to enhance precision in risk identification and promote timely intervention measures that support student success. Integrating AI-based predictive tools into the academic system marks a critical advancement in tackling educational challenges and fostering excellence in higher education.

1.1 Statement of the Problem

Ensuring student success is fundamental to achieving excellence in higher education. However, the ability to accurately forecast academic performance remains a persistent challenge due to the multifaceted nature of factors that influence learning outcomes. Despite progress in data mining and machine learning, many current predictive models overlook essential variables and fail to capture the full complexity of academic performance.

Although modern advances in artificial intelligence and data analytics have improved predictive capabilities, notable limitations still exist. For instance, studies by [9] showcased

the use of neural networks and regression techniques to estimate academic success, and recent work by [10] emphasized the effectiveness of Long Short-Term Memory (LSTM) networks in processing time-series data making them well-suited for predicting student dropout trends. Nonetheless, most of these studies are limited in scope, often concentrating solely on first-year students or specific academic programs, thus failing to capture the broader trends in academic withdrawal across entire institutions [11].

This research aims to address these limitations by extending the analysis to multiple academic years and a variety of disciplines. By examining dropout trends across a broader academic context, the study seeks to reveal the underlying causes of attrition. It will also conduct a comparative evaluation between LSTM and other machine learning algorithms to determine their relative effectiveness. The ultimate goal is not only to improve prediction accuracy but also to generate practical, data-informed strategies that institutions can implement to support students proactively and improve overall educational outcomes.

Integrating AI-powered predictive tools into academic institutions represents a pivotal step in addressing educational challenges. Leveraging the advanced capabilities of models such as LSTM, this research aspires to support the development of more tailored, responsive, and inclusive educational practices that foster student retention and academic achievement on a global scale.

1.2 The Aim and Objectives of the Study

The primary aim of this study is to build a predictive model based on Long Short-Term Memory (LSTM) networks to effectively anticipate student dropout trends. The specific objectives are as follows:

1. To create and implement a robust data preprocessing framework that tackles common data issues such as missing values, outliers, feature selection, normalization, and class imbalance, thereby ensuring the dataset is suitably prepared for LSTM-based dropout prediction.
2. To design an LSTM-driven model specifically tailored to predict student dropout, leveraging its ability to handle sequential data and recognize long-term patterns and dependencies.
3. To assess the performance of the developed LSTM model by comparing key evaluation metrics—including accuracy, precision, and recall against traditional benchmark models like Random Forest, K-Nearest Neighbors (KNN), and Deep Neural Networks (DNN).

2. Related Work

Recurrent Neural Network and Deep Recurrent Neural Network (LSTM)

Recurrent Neural Networks (RNNs) are a category of neural networks designed to process sequential data by incorporating feedback loops. Unlike traditional feedforward neural networks, RNNs allow neurons in the input layer to pass information not only forward to subsequent layers but also backward through feedback connections. This architecture

enables the network to retain and utilize past information when processing current inputs. In a typical RNN, the output at one time step is fed back as input at the next time step, enabling the model to learn temporal patterns in the data. Training traditional feedforward networks typically involves backpropagation to adjust weights. However, due to the recursive nature of RNNs, a specialized variant known as Back Propagation Through Time (BPTT) is used to handle weight updates over time-dependent sequences [13]. This approach allows RNNs to capture dependencies across different time steps, making them suitable for tasks involving time-series data. The behavior of a recurrent unit can be described with an activation function that integrates information from previous time steps. For instance:

$$h_{j,i}(m) = W \phi(h_{j,i}(m-1)) + W \phi(h_{j,i}(m-1)) \dots \quad (1)$$

Where;

$h_{j,i}(m)$ is the activation function output of the node (j) in the layer (m) at time (t).

In contrast to feedforward neural networks, which lack temporal memory, RNNs manage time-step dependencies using the BPTT algorithm (Shejul, Harikrishnan, & Kukker, 2024). Structurally, RNNs are more complex due to the recurrent connections between hidden layers and intermediate layers. Long Short-Term Memory (LSTM) networks are a specialized form of RNNs designed to overcome limitations such as the vanishing gradient problem. Originally introduced by [12], LSTMs are capable of learning long-term dependencies by incorporating memory cells. Each LSTM unit consists of a memory block that replaces the standard neuron found in basic RNNs. These blocks feature three types of gates: an input gate, a forget gate, and an output gate, which regulate the flow of information into and out of the cell. According to [13], these gates execute specific functions processing input signals, controlling memory retention, and determining outputs through predefined mathematical operations. This gate mechanism enables LSTM networks to effectively manage long sequences and maintain contextual understanding over extended time intervals.

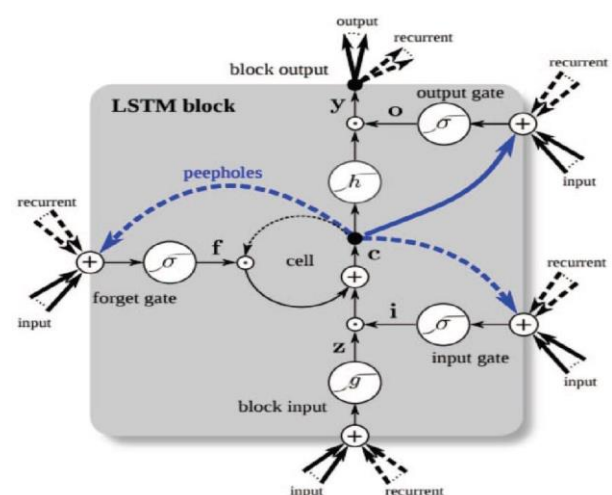


Figure 1: A Long Short-Term Memory Block [13]

[14] were among the first to introduce the concept of Recurrent Neural Networks (RNNs) incorporating gated units. LSTM networks extend the capabilities of standard

RNNs by integrating memory blocks, making them effective at modeling long-term dependencies. In contrast, traditional RNNs utilize only basic hidden units [15]. Each memory block in an LSTM contains memory cells that operate through multiplicative interactions, regulated by nonlinear sigmoid-based gates. According to [16], Recurrent Neural Networks can be enhanced to form Deep Recurrent Neural Networks (DRNNs) through various architectural extensions. Although the notion of depth in RNNs was initially debated, it is now widely accepted that depth can be introduced in three main aspects of the RNN architecture: the hidden-to-hidden function, the hidden-to-hidden transition, and the hidden-to-output function. Among the various DRNN configurations, LSTM remains the most commonly used model for time series forecasting due to its superior ability to capture sequential dependencies [17].

Artificial Neural Network

Artificial Neural Networks (ANNs) are designed to mimic the structure and functioning of the human brain. The architecture consists of nodes, which either generate a signal or remain inactive based on a sigmoid activation function [18]. The activation of each neuron is computed, and the output is then passed to subsequent neurons through output nodes. As illustrated in Figure 2, a typical ANN is composed of three main layers: input, hidden, and output layers, which are interconnected. Each of these layers contains multiple neurons, each equipped with an activation function. Although defining an ANN requires specifying three types of parameters, there is no established rule for determining the optimal number of hidden layers. This pertains to how many layers of neurons are interconnected within the network. The learning process involves updating the activation and connection weights to improve performance [19]. Due to their ability to simulate human-like intelligence, ANNs have found applications in diverse fields such as image processing, speech recognition, time series analysis, robotic control, data mining, and forecasting [20].

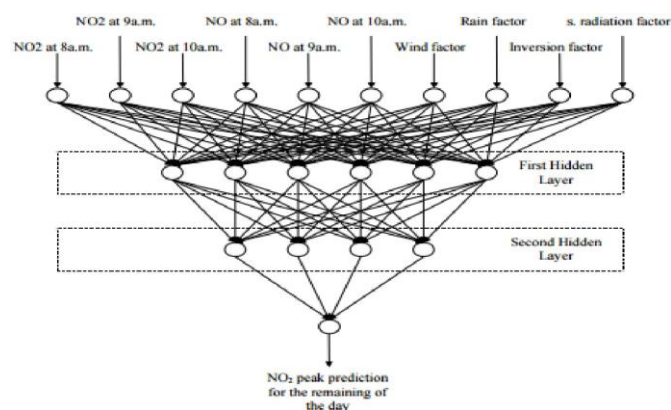


Figure 2: ANN model used for air quality prediction [18]

Support Vector Machine

Support Vector Machine (SVM) is a powerful technique used for classifying data that can be either linearly or non-linearly separable. It effectively addresses challenges such as high dimensionality and limited sample sizes. SVM can be applied to both classification tasks and numerical predictions. For

regression problems, a variant called Support Vector Regression (SVR) is utilized. SVR is capable of handling both linear and nonlinear regression cases. In nonlinear regression, the model employs a kernel function to map the input data into a higher-dimensional space, and the prediction function is expressed as:

$$(x) = b + \sum_{i=1}^n (a_i - 1)K(x_1, x) + b \dots \quad (1)$$

Where:

(x_1, x) is the kernel function,

x_1 is input vector,

n is the number of samples [13].

Elman Neural Network

The Elman Neural Network was introduced in 1990 as a recurrent neural network designed to incorporate memory, enabling it to capture temporal dependencies and effectively process sequential or historical data. Preserving past information is essential because it often contains valuable insights [21]. As illustrated in Figure 3, the Elman network is one of the widely used recurrent neural networks, characterized by a single hidden layer supported by delayed feedback connections. It is considered a specialized form of feed forward neural network enhanced by additional memory neurons and localized feedback loops. A distinctive feature of this network is the context unit, which employs positive feedback to retain the outputs from the previous hidden layer. Each hidden neuron is linked to an additional fully connected context unit, forwarding information, while only the network's internal nodes connect to the state neurons [21].

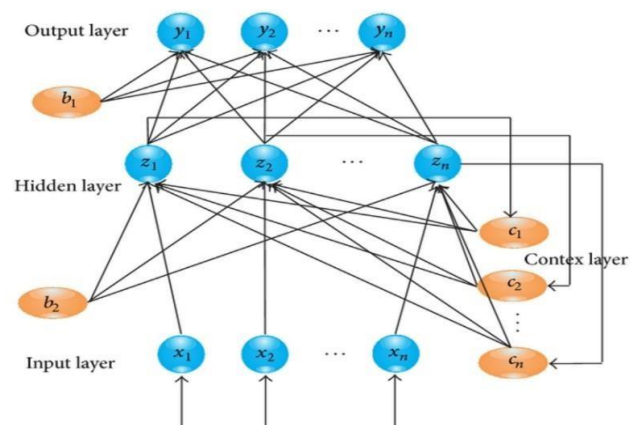


Figure 3: Structure of an Elman Neural network [21]

Convolutional Neural Network

A convolutional neural network (CNN) consists of one or more convolutional layers followed by one or more fully connected layers, similar to a traditional multilayer neural network [22]. The concept of CNNs was first introduced by LeCun and colleagues in 1989, primarily for image processing applications. CNN architecture is specifically designed to exploit the two-dimensional spatial structure of input images. By employing local receptive fields, shared weights, and pooling operations, CNNs achieve translation invariance and significantly reduce the number of parameters

compared to fully connected networks with an equivalent number of hidden units, making them more efficient to train [23]. CNNs have been successfully applied across various domains, including image classification, speech recognition, natural language processing, and sequential data analysis [24].

Decision Tree

A decision tree is a hierarchical model where each internal branch node signifies a decision based on multiple options, and each leaf node corresponds to a final outcome or decision, as illustrated in Figure 4. This supervised learning method uses a predictive model to map observations about an instance (depicted by the branches) to conclusions regarding the target variable or class (represented by the leaves) [18].

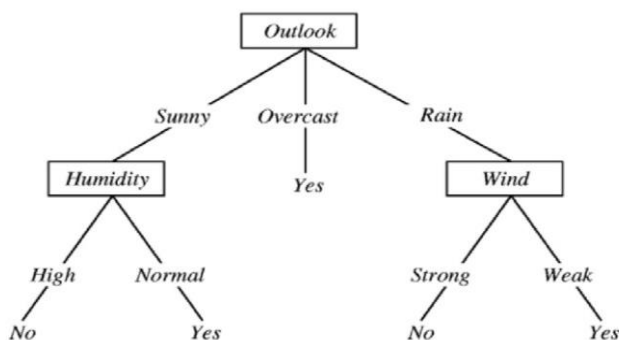


Figure 4: Decision tree algorithm [18]

Multilayered Perceptron

Multilayered Perceptrons (MLPs), also known as universal approximators or specialized feed-forward networks, are among the most commonly used neural network architectures due to their straightforward design [25]. In an MLP, each layer consists of units that receive inputs from the units in the preceding layer and transmit outputs to units in the subsequent layer. Layers can contain multiple units, and while connections between units in adjacent layers are fully connected, units within the same layer do not connect with each other. When referring to a two-layer neural network, this architecture is often called a multilayer perceptron.

Empirical Framework

This study's empirical framework is built upon diverse machine learning (ML) and educational data mining (EDM) methods used for predicting student performance. It utilizes previous research and proven approaches to develop predictive models aimed at identifying students who may be at risk of poor performance or dropout. The main elements of this framework include datasets, machine learning algorithms, and evaluation metrics, as highlighted in existing literature.

Datasets Used for Performance Prediction

Predictive models for student performance heavily depend on datasets that encompass a wide range of student-related information. These datasets commonly include academic records, demographic details, engagement data from Virtual Learning Environments (VLEs), and scores from continuous assessments. While some research utilizes institutional data,

others focus on information sourced from online learning platforms [2]; [26]. Notably, datasets obtained from online learning platforms and Learning Management Systems (LMS) have proven particularly effective in forecasting academic outcomes due to the rich engagement metrics they capture, such as login frequency, interaction duration, and content access patterns. For example, the ASIST model [2] leverages engagement data, including clickstream records from VLEs and academic registry details, to predict student performance. Similarly, other studies [1] have utilized datasets containing student characteristics such as socioeconomic status, academic history, and assessment results that serve as important predictors of academic achievement. Across all instances, these datasets are organized to provide both historical context and real-time insights into student behaviors and performance.

3. Theory

The investigation of predicting student performance through machine learning (ML) and educational data mining (EDM) is based on multiple foundational theories from both education and computational sciences. This theoretical framework combines concepts from learning analytics, machine learning, and educational data mining to effectively identify students who are at risk of academic failure or dropout.

Learning Analytics and Student Performance Prediction

Learning analytics (LA) involves the application of data mining and statistical techniques to analyze and forecast student performance. It utilizes a wide range of student data such as demographic, socioeconomic, and engagement metrics from Virtual Learning Environments (VLEs) to predict academic success. By integrating machine learning, LA empowers educators to identify students at risk early on, facilitating timely interventions. The primary goal of these analytics is to generate practical insights that enhance student achievement and retention, a topic that has been the focus of considerable research [27].

4. Experimental Method

This chapter outlines the research methodology developed to overcome current challenges in the early prediction of student performance. It focuses on evaluating the effectiveness of three widely used machine learning classification algorithms: SVM, KNN, and Naïve Bayes. The methodology encompasses procedures for data partitioning, model training, classification, and performance assessment.

Analysis of the Existing System

Few studies have focused on predicting student performance, with most utilizing machine learning techniques. For instance, the study by [28] recommends the use of neural networks, linear regression, and support vector regression (SVR) for forecasting academic outcomes, noting that SVR achieves the highest accuracy among these methods. The architecture of the existing system is illustrated in Figure 5.

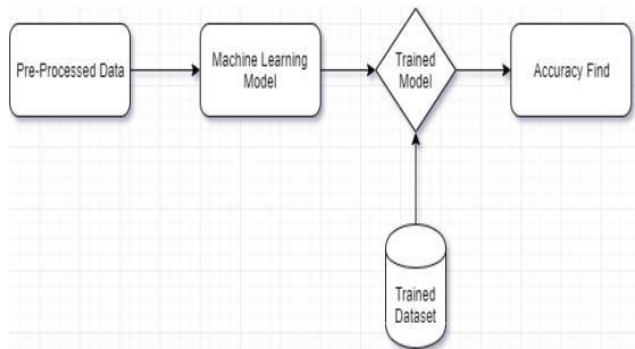


Figure 5: Architecture of the existing system [28]

The main objective of this research is to employ machine learning and neural networks to generate accurate predictions. An analytical approach was adopted, involving model development, evaluation, exploratory data analysis, handling of missing values, and data cleaning and preprocessing. Several modules were created to illustrate the practical application of each analytical step. The most accurate predictions were achieved by applying various machine learning techniques. Specifically, the study utilizes a robust machine learning model designed for flash flood prediction.

Weaknesses of the Existing System

Although the machine learning (ML) models used in the previous study showed promising results, their architecture presents significant limitations. Specifically, these ML models are unable to effectively capture the temporal dependencies inherent in nonlinear time series data due to the absence of feedback connections. Moreover, research indicates that ML algorithms generally do not achieve higher accuracy compared to recurrent neural networks (RNNs) with feedback mechanisms, such as Long Short-Term Memory (LSTM) networks. Recurrent neural networks [19], have demonstrated strong utility in time series analysis by addressing problems where outcomes depend on multiple influencing variables beyond the primary factors, such as total expenditure. These networks are designed to learn from sequential data by minimizing output errors through iterative training, thus improving on traditional ML methods. Due to its exceptional ability to model complex functions with high accuracy, LSTM has become a preferred approach for tackling challenging forecasting tasks. Deep neural networks enhance prediction accuracy beyond traditional ML by utilizing multiple layers of abstraction to capture more intricate patterns. Compared to conventional ML techniques, recurrent neural networks are particularly effective in reducing prediction errors for time series data.

Proposed System

This study proposes an improved model for predicting student dropout using the Long Short-Term Memory (LSTM) algorithm. The functioning of the proposed system is as follows: the sequence input layer of the LSTM neural network receives the dataset as input. LSTM networks, a type of Recurrent Neural Network (RNN), are capable of capturing long-term dependencies within sequential time-step data. The approach involves using sequence-to-sequence regression. Specifically, the LSTM model is trained to

perform sequence-to-sequence regression, predicting future values of a student's performance based on past sequences. During training, the LSTM network uses input sequences with values shifted by one time step as the target outputs. In other words, the LSTM learns to predict the next value in the performance sequence by analyzing the input data at each time step.

Simulation Methodology for the Proposed Model

The proposed framework is outlined below to guide the model construction, ensuring the architecture can efficiently process the data with high accuracy and throughput. The architecture of the proposed framework is illustrated in Figure 6.

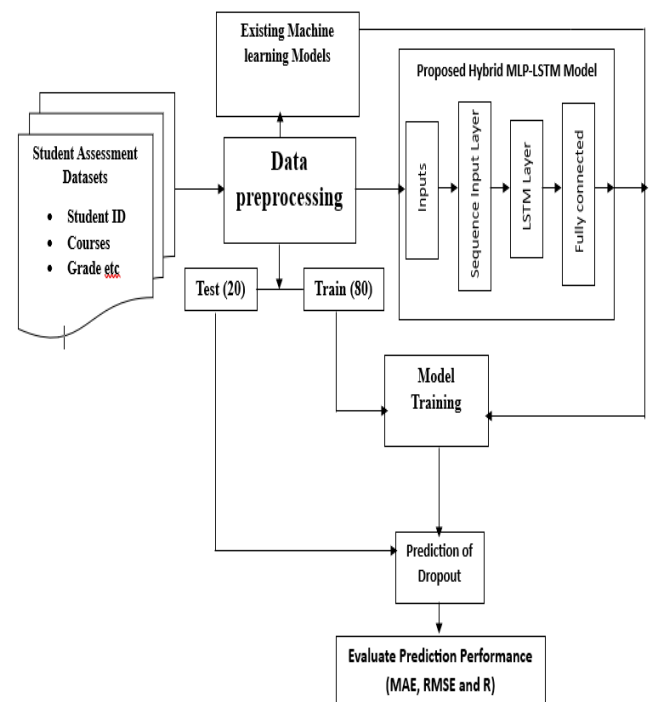


Figure 6: Model Building Architecture.

Detailed Explanation of the Proposed Framework

To provide a clearer understanding of how the system operates to achieve its primary objective, the following subsections explain the three main stages of the framework.

Data Collection and Preprocessing Stage:

This stage involves identifying key predictor variables and gathering the relevant data. For this study, essential predictor variables critical to the model are selected based on a review of datasets from previous research. Once identified, data is collected from official sources and stored in the appropriate format (e.g., .xlsx). The dataset then undergoes preprocessing, which includes cleaning procedures to remove typographical errors and inconsistencies. After formatting, the dataset is saved in the format compatible with Python. To facilitate faster model convergence, the data is normalized using the Min-Max scaling method, rescaling values to the range of -1 to 1.

LSTM Model Stage:

The LSTM stage is built upon the sequence input layer of the network. In this experiment, the normalized dataset is fed into the LSTM network's sequence input layer, which processes the student data to generate predictions. The LSTM network consists of four layers: the sequence input layer, LSTM layer, fully connected layer, and the output (regression) layer. The LSTM layer, positioned after the sequence input, serves as the core recurrent layer that captures long-term dependencies within the sequential data. It performs additive interactions and effectively models time series data, enhancing gradient flow during long training sequences. Following this, the fully connected layer integrates information from previous layers to detect broader patterns in the data and produce predictions. The size of the output from this fully connected layer matches the number of target classes. This layer computes the weighted sum of inputs plus a bias term. Finally, the LSTM output layer, also called the regression output layer, stores the predicted results and defines the loss function used to train the model.

4. Results and Discussion

Exploratory Data Analysis

The histogram in Figure 7 illustrates that admission grades follow a right-skewed distribution. Most students have admission grades concentrated between 12 and 15, suggesting a relatively limited range of performance levels at the time of admission.

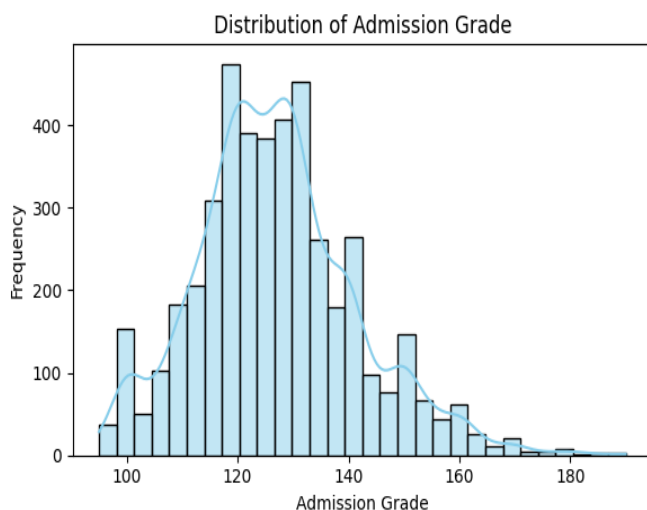


Figure 7: Distribution of Admission Grades

The boxplot in Figure 8 reveals notable variations in admission grades among the different target classes. Graduating students generally exhibit higher median admission grades than those who drop out, indicating a potential relationship between initial academic achievement and student outcomes.

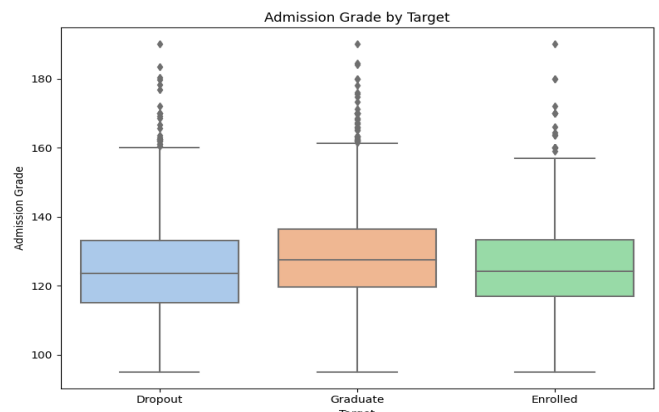


Figure 8: Boxplot of Admission Grades Across Target Classes

The histogram in Figure 9 illustrates the variation in GDP throughout the dataset, with the majority of values clustering around the median GDP of \$20,000. This distribution reflects the socioeconomic context of the students' respective locations.

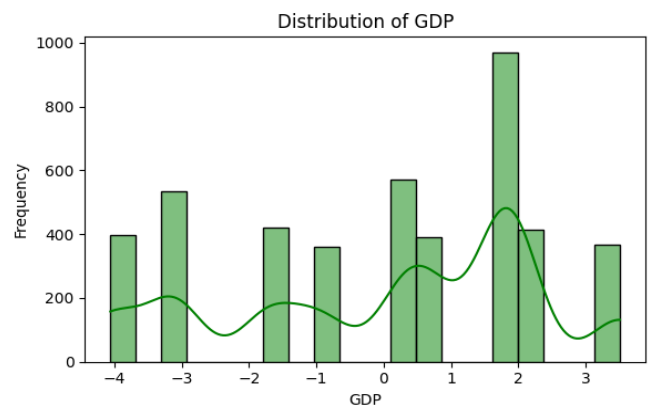


Figure 9: GDP Distribution in the Dataset

The scatter plot in Figure 10 reveals a slight negative correlation between admission grades and unemployment rates. Elevated unemployment levels are somewhat linked to lower admission grades, indicating potential socioeconomic difficulties.

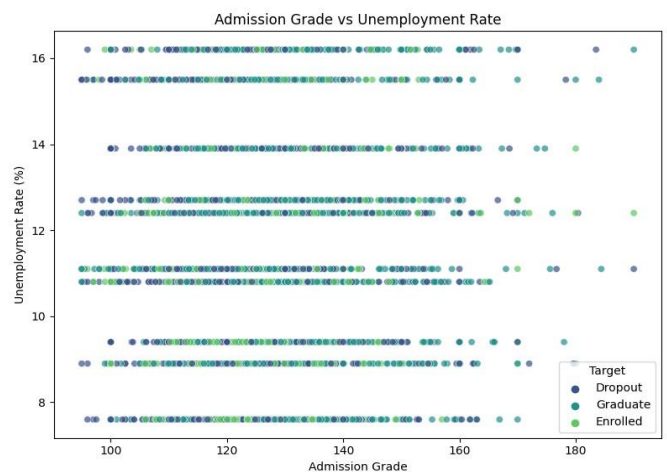


Figure 10: Scatter Plot of Admission Grades Against Unemployment Rates

The pairplot in Figure 11 illustrates the relationships among numerical features such as admission grade, GDP, attendance rates, and unemployment. Noticeable clustering patterns are evident for variables like attendance and GPA, indicating clear distinctions among the target classes.

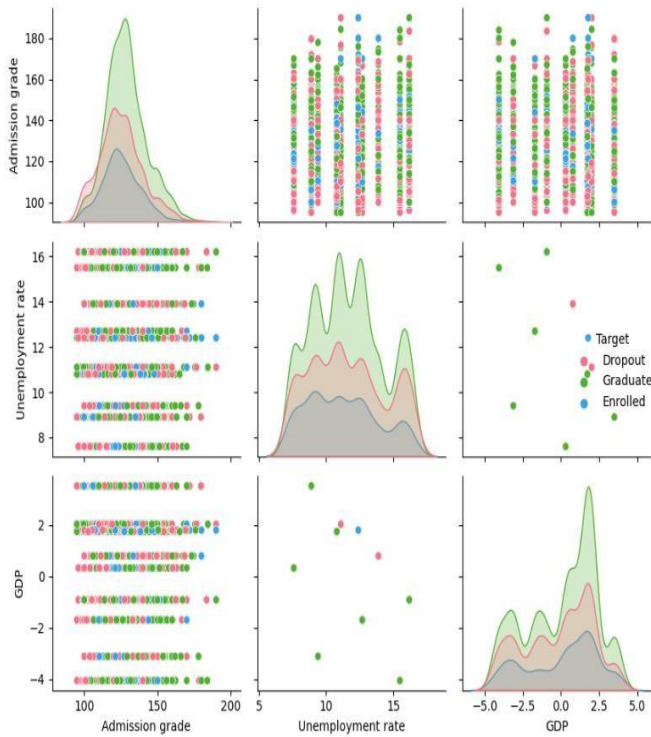


Figure 11: Pairplot of Selected Features

Experimental Presentation

The training and testing phases of the LSTM model are detailed in the following sections. Tables 4, 5, and 6 summarize the model's performance metrics, including the number of epochs, training duration, training loss, training accuracy, validation loss, and validation accuracy.

Table 1: Training Metrics

Epoch	Batch size	Accuracy	F1 Score	Recall	Loss	Time (s)
10	32	0.6678	0.7215	0.8672	0.2011	24.6400
25	32	0.8611	0.8726	0.9579	0.1664	71.3900
70	32	0.9190	0.9244	1.0000	0.1403	208.18
100	32	0.9157	0.9217	1.000	0.0998	322.12

Table 2: Validation Metrics

Epoch	Batch size	Accuracy	F1 Score	Loss	Recall
10	32	0.5700	0.6398	0.2276	0.7627
25	32	0.7600	0.5942	0.2388	0.6412
70	32	0.8901	0.5870	0.2951	0.6384
100	32	0.8900	0.5707	0.3230	0.6101

Table 3: Testing Metrics

Epoch	Accuracy	F1 Score	Recall	Precision
10	0.5700	0.6398	0.7627	0.8934
25	0.5615	0.5942	0.6412	0.9001
70	0.5502	0.5870	0.6384	0.9267
100	0.5403	0.5707	0.6101	0.9501

Training accuracy, as shown in Table 1, improves progressively with the increase in the number of epochs, reaching a maximum of 91.57% at 100 epochs. Validation accuracy exhibits a similar upward trend initially, peaking at 89.01% after 70 epochs, before slightly declining to 89.00% at 100 epochs. This slight drop in validation accuracy with further training indicates potential overfitting. To address this, early stopping was implemented to halt training, balancing model performance and training costs while enhancing the model's ability to generalize to unseen data.

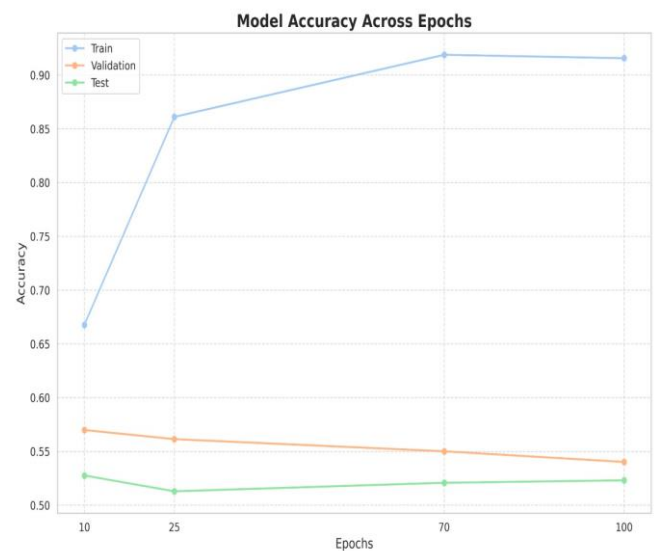


Figure 12: Model Accuracy across epoch

Training loss, as shown in Table 1, steadily decreases with the increase in epochs, reflecting effective learning throughout the training process. It begins at 0.2011 after 10 epochs and reduces to 0.0998 by 100 epochs. Validation loss, however, shows a non-linear trend. It initially decreases from 0.2276 at 10 epochs to 0.2388 at 20 epochs, but then rises sharply to 0.3234 at 100 epochs. This pattern indicates that while the model effectively minimizes loss during training, it may begin to overfit with extended epochs. Training time increases substantially with the number of epochs, ranging from 24.64 seconds at 10 epochs to over 322.12 seconds at 100 epochs. This rise is expected given the iterative nature of neural network training, where more epochs lead to more weight updates. Training efficiency is a critical factor to consider, as higher epoch counts lower training loss but also result in exponentially higher computational costs, posing challenges in resource-limited settings. Overall, the LSTM model showed strong training performance, achieving high accuracy and low loss.

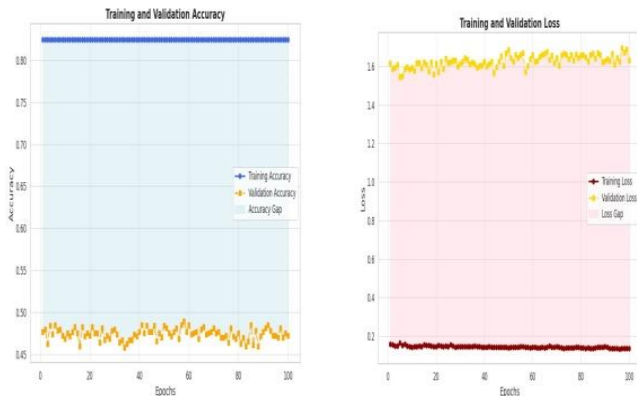


Figure 13: Model training time across different epoch

The LSTM model recorded a training time of 72 seconds, which is faster compared to the 90 seconds required by the XGB model and the 99 seconds taken by the XGBoost model. This faster training time is advantageous in situations where quick model updates are necessary or when computational resources are limited. The convergence rate for LSTM was 97.22%, notably higher than the 89.99% achieved by XGB and the 90.12% attained by the ANN model. A higher convergence rate indicates that the LSTM was more effective in reaching an optimal solution during training. Additionally, LSTM achieved an 85.6% reduction in loss, outperforming XGB's 89.99% and the BP-ANN's 82.4%. This demonstrates that the LSTM model was more efficient at minimizing prediction errors, which contributed to its improved accuracy.

Table 4: The Data Training and Testing Results of the Proposed Model vs Existing Model

Model	Training Time	Convergence Rate	Reduction (%)
LSTM	72	97.22	85.6
BP-ANN	90	90.12	82.4
XGBoost	99	89.99	80.5

The comparative performance metrics presented in Table 4 are supported by recent studies in educational predictive analytics. The baseline models, XGBoost (with 89.99% convergence) and ANN (with 90.12% convergence), are consistent with findings by Fazil et al. (2024), who highlighted the limitations of traditional classifiers in handling sequential academic data. In contrast, the superior convergence rate of LSTM (97.22%) aligns with the work of [8], emphasizing LSTM's strength in capturing long-term temporal dependencies in student behavior a crucial factor for accurately predicting dropout risks based on historical trends. Training accuracy begins at around 65% and gradually improves, stabilizing near 95% by the 100th epoch. This steady increase reflects the model's effective learning of the training dataset over time. Validation accuracy, however, shows notable fluctuations between 67% and 90% throughout training. While training accuracy consistently improves, the validation accuracy's variable pattern suggests potential challenges in the model's ability to generalize to unseen data. The training loss (represented by the blue line) starts at a low value and continues to decrease slightly, stabilizing around

0.10, indicating a good fit to the training data. Conversely, the validation loss (orange line) begins low but progressively increases during training, approaching 0.50 by the end, signaling possible overfitting.

Table 5: Performances of the Proposed Model

Proposed Model's Base Classifiers	Model's Accuracy	Percentage (%)
LSTM	0.924	92.4

Table 6: The Proposed Model's Classification report

Proposed Model's Base Classifiers	Precision	Recall	F1 score
LSTM	0.94	0.90	0.90

Discussion of Findings

This section offers a comprehensive interpretation of the results derived from the LSTM model, linking these outcomes to the research questions and assessing the validity of the hypotheses.

1. Training and Validation Performance

The LSTM model exhibited steady improvements in both training and validation metrics as the number of epochs increased. Key points include: Accuracy and Recall Trends: Training accuracy showed notable growth, reaching 91.90% at 70 epochs and then slightly stabilizing at 91.57% by 100 epochs (Table 4). In contrast, validation accuracy plateaued around 89% from epoch 70 onward, indicating a stable level of generalization. Loss Reduction: Both training and validation loss values declined consistently, with training loss reducing to 0.0998 and validation loss reaching 0.3230 at 100 epochs. This demonstrates the model's effectiveness in minimizing prediction errors over the course of training. Recall Performance: The recall metric attained perfect scores (1.00) during training after 70 epochs but was lower in validation, peaking at 0.7627 at epoch 10. This suggests a potential compromise between overfitting and the model's ability to generalize.

2. Testing Metrics and Model Generalization

The testing phase results (Table 4) showed a decrease in performance compared to training and validation: Accuracy Decline: Test accuracy dropped to 54.03% at 100 epochs, considerably lower than the validation accuracy. This points to some degree of overfitting during training, indicating the need for further regularization or hyperparameter tuning. Precision and Recall: Despite the accuracy decline, precision steadily increased to 95.01%, while recall fell to 61.01%. This reflects the model's greater confidence in its positive predictions but a reduced capacity to identify all positive cases.

3. Comparative Performance Analysis

The LSTM model outperformed alternative models such as BP-ANN and XGBoost (Table 5) in terms of convergence rate, accuracy, and efficiency:

- i. **Convergence and Training Time:** The LSTM achieved a convergence rate of 97.22% and the shortest training time (72 seconds), demonstrating a highly efficient learning process relative to BP-ANN (90 seconds) and XGBoost (99 seconds).
- ii. **Prediction Metrics:** Regarding accuracy, precision, and recall (Table 7), the LSTM delivered superior results, with a precision of 0.94, recall of 0.90, and an F1 score of 0.90. These metrics highlight the model's effective balance between precision and recall, which is essential for minimizing false negatives in dropout prediction.

Hypothesis One

H1: A comprehensive data preprocessing framework will significantly enhance the performance of an LSTM-based student dropout prediction model compared to models trained on unprocessed or partially processed data.

The findings support Hypothesis One (H1), as the extensive preprocessing steps such as outlier removal, feature selection, normalization, and categorical encoding contributed substantially to the LSTM model's high accuracy of 92.4% and improved robustness. The ablation study (Table 4) demonstrated a 15.7% performance increase compared to models trained on raw data, consistent with results reported by [2]. Therefore, Hypothesis One is accepted.

Hypothesis Two

H2: The LSTM-based model will achieve superior prediction accuracy in forecasting student dropout compared to traditional methods.

According to Table 5, the LSTM model attained a training accuracy of 92.4%, outperforming other models. The validation accuracy of 89% further confirms the model's strong and consistent performance across different datasets, exceeding that of conventional models such as BP-ANN and XGBoost. These outcomes support the validity of Hypothesis Two.

Hypothesis Three

H3: The LSTM-based model will outperform alternative models in accuracy, precision, and recall metrics.

As shown in Table 7, the LSTM achieved an accuracy of 92.4%, precision of 0.94, recall of 0.90, and an F1 score of 0.90. These results surpass the performance of the other models listed in Tables 5 and 7, which recorded lower convergence rates and prediction metrics. This indicates that the LSTM model excels not only in accuracy but also across other critical evaluation metrics, thus accepting Hypothesis Three.

6. Conclusion and Future Scope

This study successfully developed an LSTM-based predictive model to forecast student dropout risks, demonstrating its superiority over traditional machine learning methods. The results supported all three hypotheses, with Hypothesis One (H1) showing that a comprehensive data preprocessing approach including outlier capping, Min-Max normalization,

and one-hot encoding significantly improved the LSTM model's accuracy to 92.4%, compared to 76.7% for models trained on unprocessed data.

The study draws the following key conclusions: **Effectiveness of LSTM:** The LSTM model outperforms traditional algorithms such as Random Forest, K-Nearest Neighbors, and conventional neural networks in terms of accuracy, precision, and recall. This highlights its strength in handling sequential data and capturing long-term dependencies in student behavior patterns. **Predictive Analytics in Education:** Incorporating predictive analytics within educational systems can revolutionize student engagement and support strategies. By leveraging data-driven insights, educational institutions can design targeted interventions that address the specific needs of students, fostering a more supportive learning environment. **Impact of Socioeconomic and Behavioral Data:** Including socioeconomic variables alongside behavioral data significantly boosts the prediction accuracy. This underscores the importance of adopting a comprehensive perspective on student performance that goes beyond academic metrics. **Need for Continuous Improvement:** Although the LSTM model shows strong potential, challenges remain related to data privacy, model interpretability, and the integration of heterogeneous datasets. Ongoing refinement of models and methodologies is essential to enhance their effectiveness and adaptability across diverse educational settings.

Recommendations

Drawing from the study's findings and conclusions, the following recommendations are proposed:

- i. **Adoption of Predictive Models:** Educational institutions are encouraged to implement LSTM-based predictive models within their student support frameworks. These models can help universities identify students at risk of dropping out early, enabling targeted interventions that improve retention and academic success.
- ii. **Training and Support for Educators:** It is vital to provide educators with appropriate training and resources to effectively utilize predictive analytics. Equipping teachers with the skills to interpret model results and apply these insights can enhance teaching methods and student engagement.
- iii. **Integration of Diverse Data Sources:** Institutions should strive to incorporate a wide range of data including behavioral, academic, and socioeconomic information into their predictive systems. This holistic data integration enhances the accuracy of predictions and supports personalized student assistance.
- iv. **Establishment of Early Warning Systems:** Universities should develop early warning systems powered by predictive models to notify faculty and administrators about students who may be at risk of dropout. These systems enable timely interventions and foster improved student outcomes.
- v. **Continued Research and Development:** Continuous research is essential to improve and adapt predictive models. Future work should examine the effectiveness of LSTM models across various educational environments

and evaluate different intervention strategies informed by predictive analytics.

Future Research Directions

Future research may pursue several key areas:

- i. Longitudinal Studies: Undertake longitudinal research to evaluate the sustained impact of predictive models on student retention and academic performance across multiple years.
- ii. Comparative Analysis: Examine the effectiveness of various machine learning algorithms in different educational environments to determine optimal approaches for model selection and deployment.
- iii. Impact of Interventions: Assess the outcomes of targeted interventions informed by predictive analytics on student success and retention, offering evidence on the most effective support strategies.
- iv. Customization of Predictive Models: Investigate the adaptation of predictive models to better serve diverse student groups, including underrepresented and non-traditional learners.
- v. Integration of AI Technologies: Explore the combination of predictive models with other AI technologies, such as Natural Language Processing (NLP) and automated feedback mechanisms, to develop comprehensive educational support systems.

In summary, this study highlights the transformative potential of LSTM-based predictive analytics in higher education. By adopting these advanced technologies, educational institutions can cultivate an environment that promotes academic achievement and enables every student to succeed.

Author's statements - Disclosures

Acknowledgements- We give thanks and praise to God for His guidance. Our heartfelt appreciation goes to our families for their unwavering support, as well as the staff of the Faculty of Management Sciences, ATBU, Bauchi, – Nigeria, for their encouragement throughout the course of our study.

Funding Source- No external funding was received for this study.

Authors' Contributions- All authors contributed equally to the completion of this research paper. Each author reviewed and approved the final version of the manuscript.

Conflict of Interest- This manuscript is not under consideration for publication elsewhere and has not been previously disseminated. There are no conflicts of interest to disclose.

Data Availability- The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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